

APARRI, Philippines

WMO: 982320

Lat: 18.3604N

Lon: 121.6304E

Elev: 3

StdP: 101.29

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.4	21.3	18.3	13.2	21.8	19.0	13.8	22.0	12.4	22.6	11.9	22.9	5.0	40	0.394

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	6.3	33.7	28.5	33.1	28.4	32.5	28.2	29.1	32.6	28.7	32.2	28.5	31.9	3.1	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.1	24.3	31.8	27.7	23.7	31.6	27.5	23.5	31.4	94.8	32.7	93.5	32.6	92.4	32.2	32.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.8	10.2	8.9	18.7	36.6	1.6	1.6	17.5	37.8	16.5	38.7	15.6	39.7	14.5	40.8		
(5)			17.2	30.3	1.5	1.2	16.1	31.1	15.2	31.8	14.4	32.5	13.3	33.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.3	24.4	24.9	26.5	28.1	29.0	29.5	28.9	28.7	28.4	27.8	26.6	25.1	
	DBStd	2.12	1.49	1.65	1.56	1.37	1.12	0.99	1.17	1.08	0.98	1.07	1.21	1.44	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDD10.0	6330	446	417	512	544	588	584	586	580	553	551	499	468	
	CDD18.3	3288	188	184	254	294	330	334	328	322	303	293	249	209	
	CDH23.3	33740	976	1176	2222	3252	3971	4184	3941	3791	3500	3143	2303	1282	
CDH26.7	10943	72	152	515	1109	1596	1852	1568	1426	1231	881	424	117		
Wind	WSAvg	3.5	3.9	3.8	3.7	3.2	2.9	2.7	2.9	3.0	3.0	4.0	4.1	4.6	
Precipitation	PrecAvg	2111	116	71	45	33	110	130	196	218	225	319	326	205	
	PrecMax	3066	418	346	280	144	314	420	696	616	641	820	1342	496	
	PrecMin	1416	2	0	0	0	1	3	12	18	10	50	31	23	
	PrecStd	531	92	65	57	36	78	96	161	136	131	170	272	119	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.2	30.6	32.2	33.8	34.2	34.3	34.2	33.5	33.0	32.3	31.3	30.4	
		MCWB	25.9	27.2	27.6	28.4	28.6	28.8	28.6	28.5	28.5	28.3	27.7	27.1	
	2%	DB	28.2	29.4	31.0	32.5	33.1	33.6	33.1	32.7	32.1	31.4	30.3	28.9	
		MCWB	25.5	26.1	27.4	28.0	28.3	28.6	28.4	28.4	28.0	27.8	27.2	26.2	
	5%	DB	27.5	28.4	30.2	31.7	32.5	33.0	32.5	32.1	31.5	30.7	29.5	27.9	
		MCWB	25.1	25.4	26.8	27.8	28.1	28.4	28.2	28.2	27.8	27.4	26.7	25.5	
	10%	DB	26.8	27.6	29.3	31.0	31.8	32.4	31.7	31.4	30.9	30.0	28.8	27.1	
		MCWB	24.6	25.0	26.3	27.4	27.9	28.2	27.9	27.9	27.5	27.0	26.3	25.0	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.5	27.5	28.4	28.9	29.2	29.7	29.2	29.2	29.0	28.6	28.2	27.6
			MCDB	28.6	30.0	31.6	32.5	32.8	32.8	32.7	32.4	32.2	31.7	30.7	29.8
2%		WB	25.8	26.6	27.7	28.5	28.7	29.1	28.8	28.7	28.5	28.1	27.4	26.4	
		MCDB	27.8	29.1	30.6	31.9	32.3	32.7	32.4	32.0	31.6	31.0	29.9	28.4	
5%		WB	25.2	25.8	27.1	28.1	28.4	28.7	28.5	28.4	28.1	27.6	26.9	25.7	
		MCDB	27.2	28.0	29.9	31.4	31.9	32.3	31.9	31.6	31.1	30.3	29.2	27.6	
10%		WB	24.7	25.2	26.5	27.6	28.0	28.4	28.1	28.0	27.7	27.2	26.4	25.2	
		MCDB	26.5	27.2	29.1	30.6	31.5	31.9	31.4	31.1	30.5	29.7	28.5	26.9	
Mean Daily Temperature Range		5% DB	MDBR	3.8	4.2	4.9	5.6	6.0	6.3	5.8	5.4	5.3	4.5	3.9	3.3
			MCDBR	5.0	5.6	6.4	6.7	6.9	7.0	6.8	6.4	6.1	5.5	5.1	4.8
	MCWBR		3.7	4.0	4.3	4.2	4.0	4.0	3.9	3.9	3.9	3.6	3.7	3.3	
	5% WB	MCDBR	4.7	5.0	6.1	6.5	6.6	6.7	6.5	6.1	5.8	5.3	4.9	4.5	
		MCWBR	3.8	4.1	4.5	4.5	4.3	4.2	4.0	4.1	4.1	3.8	3.8	3.4	
Clear-Sky Solar Irradiance	taub	0.438	0.435	0.494	0.495	0.464	0.447	0.440	0.446	0.459	0.474	0.472	0.482		
	taud	2.325	2.352	2.175	2.201	2.330	2.391	2.409	2.381	2.341	2.263	2.252	2.181		
	Ebn at Noon	841	867	825	824	839	845	853	854	844	818	801	781		
	Edh at Noon	121	124	152	149	130	121	119	123	127	134	130	137		
All-Sky Solar Radiation	RadAvg	3.36	4.24	5.02	6.03	5.97	5.99	5.48	5.08	4.97	4.25	3.57	2.83		
	RadStd	0.33	0.62	0.49	0.47	0.40	0.47	0.50	0.41	0.38	0.33	0.48	0.39		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (3 neighbors)	N/A	N/A	N/A	-0.39	-0.33	-0.25	N/A	N/A	N/A

Nomenclature: See separate page